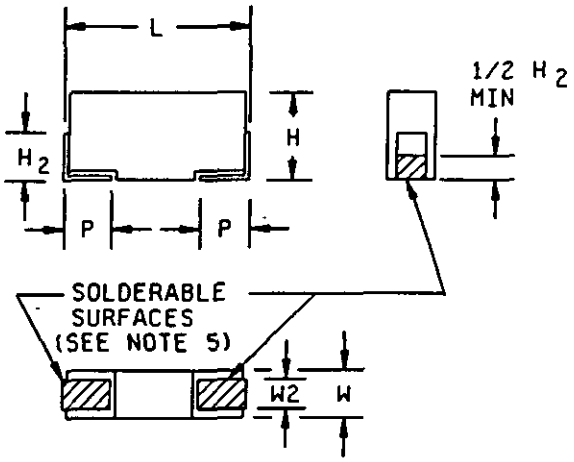
METRIC
MIL-C-55365/8C
6 October 1994
SUPERSEDING
MIL-C-55365/8B
19 November 1992

MILITARY SPECIFICATION SHEET

CAPACITORS, CHIP, FIXED, TANTALUM, ESTABLISHED RELIABILITY, STYLE CWR11 (METRIC)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the capacitors described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-55365.



Dimensions						
Case code	H	H2 (min)	L	P ±0.3 (.012)	W	W2 ±0.1 (.004) (see note 6)
A	1.6 ±.2 (.063 ±.008)	0.7 (.028)	3.2 ±.2 (.126 ±.008)	0.8 (.031)	1.6 ±.2 (.063 ±.008)	1.2 (.047)
B	1.9 ±.2 (.075 ±.008)	0.7 (.028)	3.5 ±.2 (.138 ±.008)	0.8 (.031)	2.8 ±.2 (.110 ±.008)	2.2 (.087)
C	2.5 ±.3 (.098 ±.012)	1.0 (.039)	6.0 ±.3 (.236 ±.012)	1.3 (.051)	3.2 ±.3 (.126 ±.012)	2.2 (.087)
D	2.8 ±.3 (.110 ±.012)	1.0 (.039)	7.3 ±.3 (.287 ±.012)	1.3 (.051)	4.3 ±.3 (.169 ±.012)	2.4 (.094)

FIGURE 1. Style CWR11 (molded) capacitors.

Ⓒ denotes changes

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NOTES:
1. Dimensions are in millimeters.
2. Inch equivalents are in parentheses and are given for information only.
3. These capacitors are designed for mounting by dip or wave soldering, reflow soldering, thermo-compression bonding or by other conventional means.
4. The anode (+) terminal shall be indicated by marking on the case or by bevel on anode end.
5. Solderable surfaces are only those surfaces designated as such. Termination edges are not considered solderable.
6. Applicable to solderable surfaces only. The configuration of the nonsolderable area is optional.
7. For hot solder dipped terminations (termination finish code C), add 0.38 (.015 inch) to the termination dimension tolerances.

FIGURE 1. Style CWR11 (molded) capacitors - Continued.

REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Case type: Molded chip.

Material: Sintered tantalum.

Leads: Unleaded.

Termination finish: In accordance with MIL-C-55365.

DC rated voltage: See table I. Above +85°C, voltage derating is required (see MIL-C-55365).

Operating temperature range: -55°C to +125°C.

Failure rate level: In accordance with MIL-C-55365.

DC leakage (DCL): See table I.

Capacitance (Cap.): See table I.

Cap. tolerance: ±5 percent (J), ±10 percent (K), or ±20 percent (M).

Dissipation factor (DF): See table I.

Equivalent series resistance (ESR) at 100 kHz: In accordance with MIL-C-55365. See table I.

Resistance to soldering heat: In accordance with MIL-C-55365.

Stability at low and high temperatures: In accordance with MIL-C-55365.

Step 1 (+25°C):	DCL	-	See table I
	Cap.	-	Within tolerance of table I value.
	DF	-	See table I.
	ESR	-	See table I.
Step 2 (-55°C):	ΔCap.	-	Shall not exceed ±10 percent of initial measurement.
	DF	-	See table I.
Step 3 (+25°C):	DCL	-	See table I.
	ΔCap.	-	Shall not exceed ±5 percent of initial +25°C measurement.
	DF	-	See table I.

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© TABLE I. Style CWR11 characteristics.

PIN 1/	DC rated voltage (+85°C)	Cap. (nom)	DC leakage (max)			Dissipation factor (max)			Max. ESR 100 kHz 25°C	Case code
			+25°C	+85°C	+125°C	+25°C	+85°C +125°C	-55°C		
	Volts	µF	µA	µA	µA	%	%	%	Ohms	
CWR11C-225---	4	2.2	0.5	5.0	6.0	6	9	9	8.0	A
CWR11C-475---	4	4.7	0.5	5.0	6.0	6	9	9	8.0	A
CWR11C-685---	4	6.8	0.5	5.0	6.0	6	9	9	5.5	B
CWR11C-106---	4	10.0	0.5	5.0	6.0	6	9	9	4.0	B
CWR11C-156---	4	15.0	0.6	6.0	7.2	6	9	9	3.5	B
CWR11C-336---	4	33.0	1.3	13.0	15.6	6	9	9	2.2	C
CWR11C-686---	4	68.0	2.7	27.0	32.4	6	9	9	1.1	D
CWR11C-107---	4	100.0	4.0	40.0	48.0	8	12	12	0.9	D
CWR11D-155---	6	1.5	0.5	5.0	6.0	6	9	9	8.0	A
CWR11D-225---	6	2.2	0.5	5.0	6.0	6	6	9	8.0	A
CWR11D-335---	6	3.3	0.5	5.0	6.0	6	9	9	8.0	A
CWR11D-475---	6	4.7	0.5	5.0	6.0	6	9	9	5.5	B
CWR11D-685---	6	6.8	0.5	5.0	6.0	6	6	9	4.5	B
CWR11D-106---	6	10.0	0.6	6.0	7.2	6	9	9	3.5	B
CWR11D-156---	6	15.0	0.9	9.0	10.8	6	6	9	3.0	C
CWR11D-226---	6	22.0	1.4	14.0	16.8	6	9	9	2.2	C
CWR11D-476---	6	47.0	2.8	28.0	33.6	6	6	9	1.1	D
CWR11D-686---	6	68.0	4.3	43.0	51.6	6	9	9	0.9	D
CWR11F-105---	10	1.0	0.5	5.0	6.0	4	6	6	10.0	A
CWR11F-155---	10	1.5	0.5	5.0	6.0	6	6	9	8.0	A
CWR11F-225---	10	2.2	0.5	5.0	6.0	6	9	9	8.0	A
CWR11F-335---	10	3.3	0.5	5.0	6.0	6	9	9	5.5	B
CWR11F-475---	10	4.7	0.5	5.0	6.0	6	9	9	4.5	B
CWR11F-685---	10	6.8	0.7	7.0	8.4	6	9	9	3.5	B
CWR11F-156---	10	15.0	1.5	15.0	18.0	6	6	9	2.5	C
CWR11F-336---	10	33.0	3.3	33.0	39.6	6	6	9	1.1	D
CWR11F-476---	10	47.0	4.7	47.0	56.4	6	9	9	0.9	D
CWR11H-684---	15	0.68	0.5	5.0	6.0	4	6	6	12.0	A
CWR11H-105---	15	1.0	0.5	5.0	6.0	4	6	6	10.0	A
CWR11H-155---	15	1.5	0.5	5.0	6.0	6	9	9	8.0	A
CWR11H-225---	15	2.2	0.5	5.0	6.0	6	9	9	5.5	B
CWR11H-335---	15	3.3	0.5	5.0	6.0	6	8	9	5.0	B
CWR11H-475---	15	4.7	0.7	7.0	8.4	6	9	9	4.0	B
CWR11H-106---	15	10.0	1.6	16.0	19.2	6	8	9	2.5	C
CWR11H-226---	15	22.0	3.3	33.0	39.6	6	8	9	1.1	D
CWR11H-336---	15	33.0	5.3	53.0	63.6	6	9	9	0.9	D
CWR11J-474---	20	0.47	0.5	5.0	6.0	4	6	6	14.0	A
CWR11J-684---	20	0.68	0.5	5.0	6.0	4	6	6	12.0	A
CWR11J-105---	20	1.0	0.5	5.0	6.0	4	6	6	10.0	A
CWR11J-155---	20	1.5	0.5	5.0	6.0	6	9	9	6.0	B

See footnote at end of table.

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© TABLE I. Style CWR11 characteristics - Continued.

PIN 1/	DC rated voltage (+85°C)	Cap. (nom)	DC leakage (max)			Dissipation factor (max)			Max. ESR 100 kHz 25°C	Case code
			+25°C	+85°C	+125°C	+25°C	+85°C +125°C	-55°C		
	Volts	µF	µA	µA	µA	%	%	%	Ohms	
CWR11J-225---	20	2.2	0.5	5.0	6.0	6	8	9	5.0	B
CWR11J-335---	20	3.3	0.7	7.0	8.4	6	9	9	4.0	B
CWR11J-475---	20	4.7	1.0	10.0	12.0	6	8	9	3.0	C
CWR11J-685---	20	6.8	1.4	14.0	16.8	6	9	9	2.4	C
CWR11J-156---	20	15.0	3.0	30.0	36.0	6	8	9	1.1	D
CWR11J-226---	20	22.0	4.4	44.0	52.8	6	9	9	0.9	D
CWR11K-334---	25	0.33	0.5	5.0	6.0	4	6	6	15.0	A
CWR11K-474---	25	0.47	0.5	5.0	6.0	4	6	6	14.0	A
CWR11K-684---	25	0.68	0.5	5.0	6.0	4	6	6	7.5	B
CWR11K-105---	25	1.0	0.5	5.0	6.0	4	6	6	6.5	B
CWR11K-155---	25	1.5	0.5	5.0	6.0	6	8	9	6.5	B
CWR11K-225---	25	2.2	0.6	6.0	7.2	6	9	9	3.5	C
CWR11K-335---	25	3.3	0.9	9.0	10.8	6	8	9	3.5	C
CWR11K-475---	25	4.7	1.2	12.0	14.4	6	9	9	2.5	C
CWR11K-685---	25	6.8	1.7	17.0	20.4	6	9	9	1.4	D
CWR11K-106---	25	10.0	2.5	25.0	30.0	6	8	9	1.2	D
CWR11K-156---	25	15.0	3.8	38.0	45.6	6	9	9	1.0	D
CWR11M-104---	35	0.10	0.5	5.0	6.0	4	6	6	24.0	A
CWR11M-154---	35	0.15	0.5	5.0	6.0	4	6	6	21.0	A
CWR11M-224---	35	0.22	0.5	5.0	6.0	4	6	6	18.0	A
CWR11M-334---	35	0.33	0.5	5.0	6.0	4	6	6	15.0	A
CWR11M-474---	35	0.47	0.5	5.0	6.0	4	6	6	10.0	B
CWR11M-684---	35	0.68	0.5	5.0	6.0	4	6	6	8.0	B
CWR11M-105---	35	1.0	0.5	5.0	6.0	4	6	6	6.5	B
CWR11M-155---	35	1.5	0.5	5.0	6.0	6	8	9	4.5	C
CWR11M-225---	35	2.2	0.8	8.0	9.6	6	8	9	3.5	C
CWR11M-335---	35	3.3	1.2	12.0	14.4	6	8	9	2.5	C
CWR11M-475---	35	4.7	1.7	17.0	20.4	6	8	9	1.5	D
CWR11M-685---	35	6.8	2.4	24.0	28.8	6	9	9	1.3	D
CWR11N-104---	50	0.10	0.5	5.0	6.0	4	6	6	22.0	A
CWR11N-154---	50	0.15	0.5	5.0	6.0	4	6	6	17.0	B
CWR11N-224---	50	0.22	0.5	5.0	6.0	4	6	6	14.0	B
CWR11N-334---	50	0.33	0.5	5.0	6.0	4	6	6	12.0	B
CWR11N-474---	50	0.47	0.5	5.0	6.0	4	6	6	8.0	C
CWR11N-684---	50	0.68	0.5	5.0	6.0	4	6	6	7.0	C
CWR11N-105---	50	1.0	0.5	5.0	6.0	4	6	6	6.0	C
CWR11N-155---	50	1.5	0.8	8.0	9.6	6	8	9	4.0	D
CWR11N-225---	50	2.2	1.1	11.0	13.2	6	8	9	2.5	D
CWR11N-335---	50	3.3	1.7	17.0	20.4	6	9	9	2.0	D
CWR11N-475---	50	4.7	2.4	24.0	28.8	6	9	9	1.5	D

1/ Complete PIN shall include additional symbols to indicate termination finish, capacitance tolerance, failure rate level, and if applicable, surge current option letter A (+25°C±5°C) or B (-55°C±5°C and +85°C±5°C). If optional surge current is not required, the last "-" shall be deleted.

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Step 4 (+85°C):	DCL	-	See table I.
	ΔCap.	-	Shall not exceed ±10 percent of initial +25°C measurement.
	DF	-	See table I.
Step 5 (+125°C):	DCL	-	See table I.
	ΔCap.	-	Shall not exceed ±12 percent of initial +25°C measurement.
	DF	-	See table I.
Step 6 (+25°C):	DCL	-	See table I.
	ΔCap.	-	Shall not exceed ±5 percent of initial +25°C measurement.
	DF	-	See table I.
	ESR	-	See table I.

Life: Method 108 of MIL-STD-202.

2,000 hours:

At +25°C:	DCL	-	See table I.
	ΔCap.	-	Within ±5 percent of initial measured value.
	DF	-	See table I.
	ESR	-	See table I.
At +85°C:	DCL	-	See table I.
At +125°C:	DCL	-	See table I.

(C) Surge current (optional): This test, when applicable, shall be performed on a 100 percent basis following voltage conditioning of the qualification and group A tests. Capacitors shall be subjected to ten consecutive cycles under the following conditions:

Temperature: Option A: +25°C ±5°C; option B: -55°C ±5°C and +85°C ±5°C.

Applied voltage: Rated dc voltage ±2 percent from a power source having an energy storage bank of 50,000 μf (minimum) across the output terminals.

Charge cycle: 4 seconds ±1 second.

Discharge cycle: 4 seconds ±1 second to a voltage below 1 percent of rated voltage.

The total dc resistance (excluding the capacitor) including the wiring, fixturing, and output impedance of the regulated power supply to each test position during the charging cycle shall be 1 ohm ±0.2 ohm.

On completion of the test, the capacitors shall meet the following requirements:

DCL: See table I.

Cap.: Shall meet the initial measured value.

DF: See table I.

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Marking: In accordance with MIL-C-55365, to be marked on top of case as follows: Print orientation is optional.

J	(+) polarity stripe, J = JAN
105X	- capacitance in picofarad code; manufacturer's identifier

J	(+) polarity stripe, J = JAN
105 35X	- capacitance in picofarad code - rated voltage; manufacturer's identifier

Case code A

Case codes B, C, and D

X	(+) polarity stripe, X = manufacturer's identifier
105J	- capacitance in picofarad code; J = JAN

Alternate case code A marking

At the option of the manufacturer, the lot date code may be marked on the top of the case for any case code.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Navy - AS, CG, MC, OS, SH
Air Force - 17, 19, 99
NASA - NA
DLA - ES

Preparing activity:
Army - ER

Agent:
DLA - ES

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